MODULAR CONTRACTS

Jason Roderick Donaldson Giorgia Piacentino Zichen Zhao

FACTS

Current theory

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Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Real contracts: Written in language and divided into clauses ("modules")

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Evaluated clause by clause

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- Conservatism: High burden of truth to verify clause

QUESTIONS

Why are contracts modular, with clauses evaluated separately?

What explains loopholes, complexity, landmines, incompleteness,...?

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Model of modular contracts

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E.g. true data: $\mathbf{a}_j = 110, \mathbf{a}_k = 101$

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Observations: $\hat{\mathbf{a}}_j = 110$, $\hat{\mathbf{a}}_k = 110$; conjunction: $\hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k = 110$

TRADE-OFF: WIDE VS NARROW CLAUSES

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Wide clauses

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Wide clauses: Susceptible to shuffling

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Narrow clauses

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Wide clauses: Susceptible to shuffling

Narrow clauses: Neglect joint info

RESULTS

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Extension: High thresholds $\Rightarrow$ conservatism

CONTRIBUTION

Relative to literature on contracting in language (Battigalli–Maggi 02)

- Develop framework in which clauses/evaluation matter

Relative to literature on contracting anomalies (e.g. Choi–Gulati–Scott 24)

- Develop formal model to rationalize them

MODEL

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(Atomic) Premises

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Shuffle risk captures asynchronous reporting

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Premises

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Premises: $\mathbf{a}_{\mathrm{IC}}$ = "interest coverage $\geq \kappa_{\mathrm{IC}}$"; $\mathbf{a}_{\mathrm{LR}}$ = "leverage ratio $\leq \kappa_{\mathrm{LR}}$"

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Data: Matrix of what premises hold when: E.g. $\left(\mathbf{a}_{\mathrm{IC}}, \mathbf{a}_{\mathrm{LR}}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$

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Clauses: E.g. $\mathcal{M}_{\mathrm{IC}} = \{\mathbf{a}_{\mathrm{IC}}\}$, $\mathcal{M}_{\mathrm{LR}} = \{\mathbf{a}_{\mathrm{LR}}\}, \mathcal{M}_{\mathrm{joint}} = \{\mathbf{a}_{\mathrm{IC}}, \mathbf{a}_{\mathrm{LR}}\}$

Contract: E.g. $\Phi^{\mathcal{A}} = \{\mathcal{M}_{\mathrm{IC}}, \mathcal{M}_{\mathrm{LR}}\}$ or $\Phi^{\mathcal{T}} = \{\mathcal{M}_{\mathrm{joint}}\}$

Data: Matrix of what premises hold when: E.g. $\left(\mathbf{a}_{\mathrm{IC}}, \mathbf{a}_{\mathrm{LR}}\right) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$

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Target policy: "interest coverage $\wedge$ leverage ratio"

DEFINITIONS: PREMISE- AND TARGET-BASED

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BENCHMARKS

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Evaluation reduces to standard logic; grouping of premises irrelevant

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In general: Target-based contract implements target policy: $[\![\Phi^{\mathcal{T}}]\!] = [\![\mathcal{T}]\!]$

LOOPHOLES

DEFINITION: LOOPHOLE STATES

Loophole states LH: Data realizations s.t. $[\![\Phi^{\mathcal{A}}]\!] \neq [\![\Phi^{\mathcal{T}}]\!]$ even if $e = 0$

EXAMPLE: LOOPHOLE

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Two premises with three data each and $\tau = 2$:

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i
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2
3

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Loophole: Evaluation & conjunction don't commute: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k]\!]$

SHUFFLE RISK

DEFINITION: SHUFFLE RISK STATES

Shuffle risk states SR: Data realizations s.t. $0 < \mathsf{Pr}\Big[\big[\!\big[\Phi^{\mathcal{T}}\big]\!\big] = 1 \,\Big|\, \{\mathbf{a}_j\}_j\Big] < 1$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
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# of 1s $\geq \tau$	**1**	**1**	**0**

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

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Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

Joint changed: $101 \rightarrow 110 \Rightarrow$ target-based evaluation flips

RESULTS

LEMMA

LEMMA: LOOPHOLES ARE FALSE POSITIVES

If $[\![\Phi^{\mathcal{A}}]\!] \neq [\![\mathcal{T}]\!]$, then $[\![\Phi^{\mathcal{A}}]\!] = 1$

LEMMA: LOOPHOLES FALSE POS: INTUITION

Harder to satisfy $\mathbf{a}_j$ and $\mathbf{a}_k$ than either individually

R1: LOOPHOLES

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With $e > 0$, if $\mathsf{Pr}[\mathsf{SR} \setminus \mathsf{LH}]$ large, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

If loopholes are infrequent, make clauses narrow

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If loopholes are infrequent, make clauses narrow

Accept loopholes to reduce shuffling error

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J. Crew was distressed and wanted to free up collateral

Credit agreement ostensibly banned operations exploiting creditors, but...

Clause $\mathbf{a}_j$: "Can move assets to restricted sub"

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I.e. loophole: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k]\!]$

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Katz 10: Many loopholes openly exploited after much litigation

Refutes form-vs-substance, intent-to-evade, spirit-vs-letter theories

Says loopholes akin to voting paradoxes (Arrow's Theorem)

Doctrinal Paradox (DP): Vote on premises $\neq$ on targets (List–Pettit 02)

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We nest DP (observations $\leftrightarrow$ agent beliefs) and connect with loopholes

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- Show why loopholes persist $e > 0$

R2: ERRORS DUE TO COMPLEXITY

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If $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

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Empirically: Complexity leads to errors in taxes and regulatory compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if there's $\mathbf{a}_j$ s.t. $\mathbf{a}_j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for clauses $m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. premise-based: Keep joint information

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Cost: False negatives

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$\Phi^{\mathcal{I}}$ called incomplete w.r.t. $\mathcal{T}$ if $\mathbf{a}_j \notin \bigcup_{\mathcal{M}_m \in \Phi^{\mathcal{I}}} \mathcal{M}_m$ for some $\mathbf{a}_j \in \mathcal{T}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\left[\!\left[\Phi^{\mathcal{I}}\right]\!\right] > \left[\!\left[\mathcal{T}\right]\!\right]\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\left[\!\left[\Phi^{\mathcal{I}}\right]\!\right] > \left[\!\left[\mathcal{T}\right]\!\right]\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

"Sometimes" also $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{A}}$

R4: INCOMPLETENESS: INTUITION

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Akin to mismeasurement in Holmström–Milgrom:

R4: INCOMPLETENESS: INTUITION

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Benefit w.r.t. premise-based: If "proxy task" $\mathbf{a}_\ell \approx \mathbf{a}_j \wedge \mathbf{a}_k$: Keep joint info

Akin to mismeasurement in Holmström–Milgrom:

Decrease power on distortionary tasks via omission or proxy

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is conservative if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM (INFORMALLY)

"Often" $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{A}}$ and $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{T}}$

R5: CONSERVATISM: INTUITION

Premise-based gives false positives; conservatism decreases them

R5: CONSERVATISM: INTUITION

Premise-based gives false positives; conservatism decreases them

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

R5: CONSERVATISM: INTUITION

Premise-based gives false positives; conservatism decreases them

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

Bug (lite): Still neglects joint info (but fewer false positives than $\Phi^{\mathcal{A}}$)

R5: CONSERVATISM: APPLICATIONS

Law: "beyond reasonable doubt," "reasonable suspicion," "probable cause"
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

CONCLUSION

Model of modular contracts in formal language s.t. shuffled data

- Loopholes
- Errors of complexity
- Landmines
- Incompleteness
- Conservatism

MODULAR CONTRACTS